

$$\text{fogStart} = 10, \quad \text{fogRange} = 200$$

when $\text{dist} = 160$,

110 60 30

$$s = \text{saturate} \left(\frac{160 - 10}{200} \right) = \frac{3}{4} = 0.75$$

$$\text{foggedColor} = (0.25) \text{litColor} + (0.75) \text{fogColor}.$$

when $\text{dist} = 110$,

$$s = \text{saturate} \left(\frac{110 - 10}{200} \right) = 0.5$$

$$\text{foggedColor} = (0.5) \text{litColor} + (0.5) \text{fogColor}.$$

when $\text{dist} = 60$,

$$s = \text{saturate} \left(\frac{60 - 10}{200} \right) = 0.25$$

$$\text{foggedColor} = (0.75) \text{litColor} + (0.25) \text{fogColor}.$$

when $\text{dist} = 30$,

$$s = \text{saturate} \left(\frac{30 - 10}{200} \right) = 0.1$$

$$\text{foggedColor} = (0.9) \text{litColor} + (0.1) \text{fogColor}.$$

$\frac{\partial P}{\partial u}, \frac{\partial P}{\partial v}$ 는 텍스처 공간에서 각각 변화가 표면 위 3D 위치를 어떤 방향으로 움직이는지 나타내는 접선 벡터이다.